

CAMPUS ASSIST BOT

Rule-Based Python Chatbot for Student Assistance

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TASK OBJECTIVE

Build a simple conversational chatbot in Python that asks for the user's name, runs continuously, recognizes greetings and student-help topics, handles unknown messages, and exits when the user enters bye or exit.

TIME PLAN

Time	Task
0–5 min	Plan the chatbot and decide what student-assistance topics it should handle.
5–20 min	Write and test the chatbot.
20–25 min	Add extra responses and test different inputs.
25–30 min	Clean the code and prepare a quick demonstration.

REQUIREMENTS

No.	Requirement
1	Ask the user's name before the main conversation.
2	Use while True: to keep the conversation running.
3	Use input() to receive messages.
4	Clean the input using .lower().strip().
5	Recognize at least two greetings, such as hello and hi.
6	Answer at least three different student questions or topics.
7	Use if / elif / else for decision making.
8	Use break to exit when the user types bye or exit.
9	Use the user's name in at least one response.
10	Provide a useful response for unknown messages using else.
11	Use the in operator for flexible greetings such as hello bot.

Challenge: Use the in operator so variations such as “hello bot” or “hey hello” can still trigger a greeting.

Theme: Student assistance – attendance, timetable, library, clubs, and campus guidance.

1. PYTHON CODE

Complete runnable code for the revised Campus Assist Bot example.

```
# Campus Assist Bot - Rule-Based Python Chatbot
print("=" * 50)
print("      CAMPUS ASSIST BOT")
print("      Student Support Assistant")
print("=" * 50)

name = input("Enter your name: ").strip()
print(f"\nWelcome, {name}!")
print("Ask me about:")
print("- Attendance")
print("- Timetable")
print("- Library")
print("- Clubs")
print("Type 'bye' or 'exit' to stop.\n")

while True:
    message = input(f"{name}: ").strip().lower()

    if message == "bye" or message == "exit":
        print(f"Bot: Goodbye, {name}! Have a great day!")
        break
    elif "hello" in message or "hi" in message or "hey" in message:
        print(f"Bot: Hello {name}! How can I help you today?")
    elif "attendance" in message or "present" in message:
        print("Bot: Check your current attendance in the student portal and "
              "contact your faculty if you find an error.")
    elif "timetable" in message or "schedule" in message:
        print("Bot: Check the latest class timetable for your section "
              "before planning your day.")
    elif "library" in message or "book" in message:
        print("Bot: The library can help you find textbooks, references, "
              "and other academic resources.")
    elif "club" in message or "event" in message:
        print("Bot: Explore student clubs and campus events to join activities "
              "that match your interests.")
    else:
        print("Bot: I can help with attendance, timetable, library, or clubs.")
```

GENERAL OUTPUT

=====

CAMPUS ASSIST BOT
Student Support Assistant

=====

Enter your name: VEMULA VISHNU

Welcome, VEMULA VISHNU!

Ask me about:

- Attendance
- Timetable
- Library
- Clubs

Type 'bye' or 'exit' to stop.

VEMULA VISHNU: hello bot

Bot: Hello VEMULA VISHNU! How can I help you today?

VEMULA VISHNU: attendance

Bot: Check your current attendance in the student portal and contact your faculty if you find an error.

VEMULA VISHNU: timetable

Bot: Check the latest class timetable for your section before planning your day.

VEMULA VISHNU: library

Bot: The library can help you find textbooks, references, and other academic resources.

VEMULA VISHNU: bye

Bot: Goodbye, VEMULA VISHNU! Have a great day!

The two screenshots from the previous version have been removed. A clean general output is provided above instead.

2. RESULT

The test cases below cover the required chatbot behaviours for the revised example.

Test Input	Expected Result	Observed Result	Status
hello	Greeting response	Greeting displayed	PASS
HELLO	Case-insensitive greeting	Greeting displayed	PASS
hello bot	Flexible greeting using in	Greeting displayed	PASS
hi	Greeting response	Greeting displayed	PASS
ATTENDANCE	Attendance guidance	Attendance advice displayed	PASS
TIMETABLE	Timetable guidance	Schedule advice displayed	PASS
LIBRARY	Library guidance	Library advice displayed	PASS
CLUBS	Club/event guidance	Club advice displayed	PASS
random message	Useful fallback using else	Fallback displayed	PASS
bye	Exit message and break	Chatbot stopped	PASS

FINAL VERIFICATION

Check	Verification
Input	User name and chatbot messages are accepted using input().
Processing	Messages are normalized with .lower().strip() and evaluated using if / elif / else.
Response	The bot provides topic-specific replies for attendance, timetable, library, and clubs.
Flexible greeting	The in operator recognizes greeting variations such as “hello bot”.
Repeat	while True keeps the conversation active until an exit command is entered.
Exit	bye or exit prints a closing message and uses break to stop the loop.
Fallback	Unknown messages receive a useful response through else.

CHATBOT FLOW

INPUT → PROCESS → RESPOND → REPEAT → EXIT

Submission Details

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